

An Adaptive Entropy-Driven Multi-Channel Blockchain Steganographic Protocol with Post-Quantum Security for Covert Communication

Enhancing Capacity, Undetectability, and Quantum Resilience in Blockchain-Based Covert Channels

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Abstract

Blockchain steganography has drawn considerable research attention as a means of achieving covert communication through decentralized infrastructure, yet several stubborn limitations persist across existing approaches. In this paper, we introduce an Adaptive Entropy-Driven Multi-Channel Blockchain Steganographic Protocol (AED-MCBSP) that seeks to push beyond these limitations through three interconnected contributions. At its core lies a Dynamic Entropy-Driven Capacity Optimization (DED-CO) model that, rather than fixing payload sizes ahead of time, continuously adjusts embedding capacity in response to observed blockchain network entropy. In practice, this yields approximately 156 bits per transaction under typical conditions — a substantial gain over the roughly 82 bits achievable through conventional fixed-capacity schemes. We also propose a Lattice-Enhanced Hierarchical Deterministic Wallet (LE-HDW) that strengthens key derivation using Ring-LWE hardness assumptions, giving the protocol post-quantum resilience without sacrificing compatibility with existing address formats. Finally, we formalize security through what we call the MASS (Multichannel Adaptive Steganographic Security) framework, supplying a game-based indistinguishability proof that bounds adversarial advantage under standard lattice assumptions. Experiments conducted on Bitcoin and Ethereum testnets suggest that AED-MCBSP simultaneously improves channel capacity, lowers computational overhead to around 6.7ms per embedding operation, and reduces transaction costs relative to prior work. We argue these qualities make the protocol particularly attractive for sensitive application domains such as healthcare data exchange and government communication channels.

Keywords:Blockchain steganography, covert communication, adaptive capacity optimization, post-quantum cryptography, lattice-based security, hierarchical deterministic wallets, entropy-driven models.

1. Introduction

Not all secure communication is created equal. Cryptography has served us remarkably well for protecting message content, but it carries an inherent visibility cost — the mere act of encrypting signals that something worth hiding is being transmitted. This encryption visibility problem has long troubled practitioners in sensitive domains where the existence of communication itself must remain secret [1]. Steganography offers a different philosophy entirely: hide the message so thoroughly that outsiders cannot even tell a conversation took place [2]. It is an old idea, but one that has gained fresh relevance as surveillance capabilities have expanded.

Blockchain technology, somewhat unexpectedly, turns out to be an unusually capable carrier for steganographic communication. The reasoning is not immediately obvious, but it becomes clearer on reflection. Blockchains process millions of routine transactions daily — payments, transfers, smart contract interactions — creating a vast stream of seemingly mundane data in which covert messages can, in principle, disappear entirely [3]. The decentralized nature means no single authority controls the channel, and the immutable ledger ensures that once embedded, messages resist tampering. These properties have motivated a growing line of research seeking to harness blockchain infrastructure for covert data transmission.

Yet existing approaches, despite genuine ingenuity, leave important gaps. Torki and colleagues showed that Hierarchical Deterministic Wallet (HDW) algorithms could embed roughly 82 bits per transaction by carefully selecting transaction output addresses [3]. Chen et al. pushed in a different direction, using reversible Generative Adversarial Networks to synthesize transaction fields with embedded payloads, achieving about 40 bits per field [4]. More recently, Omego and Bosy developed hybrid multichannel protocols with provable security guarantees, though their construction assumes fixed channel configurations [5]. These advances, while significant, share a common limitation that has not received adequate attention: they all assume embedding capacity can be determined ahead of time, independent of actual network conditions.

This assumption matters more than it might first appear. Blockchain networks are far from static. Transaction volumes fluctuate dramatically — compare a quiet Sunday morning to a Monday during a market crash. Fee structures change. Network entropy, which roughly captures how unpredictable the transaction stream appears, varies continuously. Embedding the same payload size regardless of these conditions seems, in retrospect, almost certainly suboptimal. During high-entropy periods, when the transaction stream is naturally chaotic, larger payloads could plausibly hide in plain sight. During quieter periods, conservative embedding seems the wiser course. No existing framework captures this intuition formally.

Beyond the capacity question, there is the quantum elephant in the room. Current blockchain steganographic protocols overwhelmingly rely on elliptic curve cryptography and the hardness of the Elliptic Curve Discrete Logarithm Problem (ECDLP) [6]. NIST's recent standardization of post-quantum algorithms — ML-KEM and ML-DSA, published as FIPS 203 and 204 in August 2024 — signals a broader transition that blockchain-based systems cannot afford to ignore [6]. The question is not whether quantum-resistant steganography will be needed, but when, and whether protocols being designed today will survive that transition.

This paper takes up both challenges. We present the Adaptive Entropy-Driven Multi-Channel Blockchain Steganographic Protocol (AED-MCBSP), organized around a simple but powerful idea: embedding capacity should respond to network conditions in real time, and the cryptographic foundations should withstand quantum attacks. Our DED-CO model formalizes this intuition, expressing optimal capacity as a function of measured blockchain network entropy. The LE-HDW construction replaces traditional ECDLP-based key derivation with Ring-LWE hardness assumptions, providing post-quantum security while preserving the hierarchical key management that makes HDW-based steganography practical. We prove security within the MASS framework, showing that distinguishing stego transactions from legitimate ones reduces to solving hard lattice problems. Experimental results on public testnets suggest the approach is not merely theoretically sound but practically viable, improving on existing methods across capacity, efficiency, and cost dimensions simultaneously.

A second motivation is equally important: covert communication systems are increasingly judged not only by whether they can hide information, but by whether they can continue to do so under changing operational conditions. A blockchain network is a living environment. Its transaction rhythm, fee pressure, address diversity, and field-level randomness shift from hour to hour. A protocol that treats this environment as static may perform well in controlled experiments, but it becomes fragile when the background traffic changes. AED-MCBSP is therefore designed with adaptation as a first-class principle. It does not simply embed data into blockchain transactions; it observes the network, estimates how much cover the network can safely provide, and adjusts its behaviour before a detectable pattern begins to form.

2. Background and Related Work

2.1 Early Work in Blockchain Steganography

The idea of using blockchain for covert communication did not emerge from nowhere, though its intellectual lineage is somewhat scattered. Torki et al. provided what many consider the foundational treatment, systematically identifying why blockchain makes sense as a steganographic carrier and proposing HDW-based embedding algorithms to realize the idea [3]. Their insight was elegant: rather than modifying existing blockchain data (which leaves traces), generate new transactions whose fields naturally encode the desired payload. Child addresses derived through the deterministic relationship produce outputs that are computationally indistinguishable from ordinary addresses. The medium-capacity variant achieves $nm + \log_2(n!)$ bits per transaction by embedding m bits across each of n output addresses and leveraging permutations for additional information capacity. It is clever, and it works — up to a point.

That point, in our view, is reached when one asks whether 82 bits per transaction represents an inherent limit or merely a local optimum. Partala's approach took a different tack, embedding messages directly into Bitcoin addresses through brute-force search, reportedly achieving comparable capacity but at staggering computational cost — around \$1,100 in transaction fees per message block [7]. Hardly practical for most scenarios. Zhang and Zhang experimented with OP_RETURN commands, but the low baseline usage of this opcode makes such transactions stand out to any analyst looking for anomalies [8]. Gimenez-Aguilar et al. explored Ethereum-specific features and reported impressive per-field capacity, yet their approach does not generalize across blockchain platforms [9]. In short, early work established feasibility but left substantial room for improvement on multiple fronts.

2.2 Generative Approaches and Their Limits

Chen and colleagues took the field in a markedly different direction with their Generic Blockchain-based Steganography Framework (GBSF), employing what they termed a Backcalculating Generative Adversarial Network, or CCR-GAN [4]. The architecture is conceptually appealing: a generator network maps latent noise — into which the covert payload is woven — directly to synthetic transaction fields. The backcalculation mechanism enables recovery at the receiver without storing additional metadata. Custom techniques including Counter-Intuitive Data Preprocessing (CIDP) and a specialized ClipSigmoid activation function push capacity to roughly 40 bits per field.

We find this work genuinely thought-provoking, but several concerns temper our enthusiasm. First, the computational overhead is non-trivial — the authors report 105 GPU-hours for training and 125 milliseconds per inference operation. For applications requiring real-time or near-real-time communication, this latency is difficult to accept. Second, and perhaps more fundamentally, the mapping from continuous generator outputs to discrete blockchain fields introduces rounding errors that, however small, create statistical artifacts. An adversary with sufficient transaction data might eventually learn to detect these artifacts. Third, the capacity-concealment trade-off mechanism operates statically; it does not respond to changing network conditions, which brings us back to the same rigidity problem identified in HDW-based approaches. Finally, like virtually all prior work, the

cryptographic foundations remain anchored to ECC assumptions that quantum computing threatens to undermine.

2.3 Hybrid Protocols: Where We Stand Today

The most recent advance comes from Omego and Bosy, who proposed combining Cover Modification (CMO) and Cover Synthesis (CSY) techniques within a single multichannel protocol backed by formal security analysis [5]. Their construction uses secret-seeded pseudorandom number generators driving Markov chain generators for cover parameter synthesis, together with variance-aware LSB embedding. Security is analyzed under the MC-ATTACK model, and the authors demonstrate negligible distinguishing advantage for computationally bounded adversaries.

This represents genuine progress, particularly in bringing rigorous proof techniques to the blockchain steganography literature. Nevertheless, we see two important gaps. The protocol configuration — which channels to use, at what capacity — remains fixed after initialization. There is no mechanism for responding to the kinds of network dynamics we have described. Moreover, the security analysis, while formally satisfying, assumes classical computational hardness; nothing in the construction or proof addresses quantum adversaries. Given the timeline of quantum computing development and the NIST standardization process already underway, this seems like an omission worth correcting.

3. Preliminaries

3.1 Hierarchical Deterministic Wallets

Bitcoin Improvement Proposal 32 (BIP-32) introduced Hierarchical Deterministic Wallets to solve a practical problem: how to generate arbitrarily many key pairs from a single seed while maintaining recoverability and structure [10]. The construction has since become foundational to virtually all cryptocurrency wallet software. Formally, an HDW instance H comprises a master chain code k , a master private key y , the elliptic curve generator G , and a hash function H producing 1-bit outputs. Child key derivation follows $x_i = y + H(k \parallel i) \bmod q$, with public keys computed as $P_i = x_i * G$.

Two derivation modes exist. Normal derivation requires the parent private key and produces child keys usable for signing. Hardened derivation, in contrast, uses only the parent public key and chain code, enabling so-called watch-only wallets that can monitor addresses without holding signing keys. For steganographic purposes, the deterministic relationship between index i and derived address P_i is the crucial property — it allows the sender and receiver to independently generate the same sequence of addresses from shared master parameters, creating a covert channel whose very existence is difficult to establish.

3.2 Lattice-Based Cryptography and Ring-LWE

The security of lattice-based cryptography rests on problems believed to resist both classical and quantum attacks — a property that has driven intense research interest and, ultimately, NIST standardization. The Ring Learning With Errors (Ring-LWE) problem, introduced by Lyubashevsky, Peikert, and Regev, serves as our primary hardness assumption [11].

Consider the ring $R = \mathbb{Z}[x]/(x^n + 1)$, which is the ring of integers of the $2n$ -th cyclotomic field, and its quotient $R_q = R/qR$. For a fixed secret s in R_q and an error distribution χ over R , the Ring-LWE distribution $A_{\{s, \chi\}}$ produces samples (a, b) where a is drawn uniformly from R_q , the error e is drawn from χ , and $b = a * s + e$ (all operations in R_q). The Decision Ring-LWE problem asks an adversary to distinguish polynomially many samples from $A_{\{s, \chi\}}$ from the same number of uniformly random samples over $R_q * R_q$.

The relevance to post-quantum security comes from a celebrated reduction: solving Ring-LWE is at least as hard as approximating the Shortest Vector Problem (SVP) on ideal lattices to within polynomial factors, and no quantum algorithm is known that achieves this efficiently [11]. This gives us confidence — not absolute certainty, but confidence grounded in substantial cryptanalytic effort — that constructions based on Ring-LWE will remain secure even against quantum adversaries.

3.3 Information-Theoretic Foundations

Understanding the fundamental limits of covert communication requires information-theoretic machinery. Following Bash, Goeckel, and Towsley, we model the steganographic channel as a state-dependent system characterized by the conditional distribution $p(y | x, s)$, where x denotes the covert input, y the resulting transaction (the stego output), and s the channel state capturing current network conditions [12].

For memoryless channels with independent and identically distributed states S known causally at the transmitter, capacity takes the form $C = \max I(T; Y)$, where the augmented input alphabet encodes state-contingent transmission strategies and the transition probabilities marginalize over the state distribution. This framework, developed in the context of AWGN channels, extends naturally to blockchain steganography where the channel state corresponds to observable network properties such as transaction volume, fee distributions, and timing patterns.

A long-standing result known as the square root law of steganography imposes a seemingly harsh limitation: for cover sequences of length n , the maximum covert message length scales as $O(\sqrt{n})$ if strict undetectability is required [13]. However, subsequent work by Wang, Wornell, and Zheng showed that this limitation can be circumvented when the transmitter has accurate channel state information [14]. This finding provides theoretical justification for our entropy-driven adaptive approach: by measuring and responding to network conditions, we effectively exploit channel state knowledge to operate beyond the square-root regime.

4. The AED-MCBSP Framework

4.1 Architecture Overview

AED-MCBSP is structured as four interacting modules, each addressing a specific functional requirement. Rather than describing these in isolation, it is perhaps more useful to think about what each module contributes to the overall protocol behavior and how they fit together.

Module	Purpose	Key Output
DED-CO	Adapts payload size to current network conditions	Optimal embedding capacity $C^*(t)$
LE-HDW	Derives address chains with quantum-resistant security	Lattice-enhanced key hierarchy
ATFS	Selects which transaction fields to embed within	Field selection distribution
MCDP	Distributes payload fragments across transactions	Sequence of stego transactions

The architecture layers these modules naturally. Above sits the application layer handling message segmentation, compression, and session state. The cryptographic layer implements LE-HDW key derivation together with ML-KEM-based session key establishment. Below that, the steganographic layer runs DED-CO capacity computation and ATFS field selection in tandem — the output of one informs the other. Finally, the blockchain interface layer handles the practicalities of transaction creation, signing, broadcasting to the peer-to-peer network, and monitoring for confirmation. This modularity is not merely aesthetic; it means improvements to, say, the field selection strategy can be incorporated without touching the key derivation logic, and vice versa.

The design choice behind this layered structure is intentionally conservative. In sensitive communication systems, architectural elegance matters less than operational predictability. Each module is therefore responsible for a narrow and auditable function: one estimates capacity, another derives keys, another chooses fields, and another coordinates distribution. This separation reduces hidden coupling between cryptographic and steganographic decisions. It also makes the protocol easier to review, tune, and deploy in environments where compliance teams, security architects, and application owners may each examine different parts of the system. In practical terms, the protocol can evolve without forcing a complete redesign whenever a better entropy estimator, field-risk classifier, or post-quantum primitive becomes available.

4.2 Dynamic Capacity Optimization

The DED-CO model embodies what we consider the central insight of this work: embedding capacity ought to be a function of what the network is actually doing, not a predetermined constant. To make this precise, we need a measure of network activity that correlates with how much cover the transaction stream provides.

We define the Blockchain Network Entropy as follows. Over a sliding window of Δt blocks, let $f(v)$ denote the empirical frequency distribution of transaction values v appearing in the observed

transaction set. The entropy itself combines Shannon entropy of the value distribution with a normalization factor capturing transaction timing irregularity:

$$H_{net}(t) = -\sum f(v) \log_2 f(v) * (1 - \sigma/\mu)$$

Here μ and σ are the mean and standard deviation of inter-arrival times between consecutive transactions. The normalization term $(1 - \sigma/\mu)$ has a simple interpretation: when transactions arrive regularly (low relative variance), this factor approaches one; when arrivals are bursty and unpredictable, it shrinks, reflecting reduced temporal cover. We have found this works better in practice than using Shannon entropy alone, which does not capture timing structure.

Given this entropy measure, the optimal embedding capacity becomes proportional to $H_{net}(t)$ multiplied by the base capacity from the HDW construction:

$$C_{opt}(t) = \alpha * H_{net}(t) * \sqrt{nm + \log_2(n!)}$$

The concealment parameter α in $(0, 1]$ deserves some discussion. We compute it dynamically as $\alpha = \max(0.1, 1 - \epsilon_d)$, where ϵ_d is the estimated detection probability produced by the ATFS module. This creates a feedback loop: when field selection suggests higher detection risk, capacity automatically scales back. The base term under the square root generalizes Torki et al.'s formula with square-root scaling consistent with covert communication theory [13]. The multiplicative entropy factor $H_{net}(t)$ is where the adaptation happens — capacity grows when the network is noisier and contracts when it is quieter.

In Algorithm 1, we specify the complete capacity computation procedure. The input includes current blockchain state, a maximum concealment threshold δ_{max} (we typically use 0.05), and HDW parameters (n, m) . After sampling recent transactions and computing the value frequency distribution, the algorithm evaluates Equation (1), queries the ATFS module for a detection risk estimate, computes α , evaluates Equation (2), and applies a hard cap C_{max} as a safety mechanism. The capped output $C^*(t)$ is what downstream modules actually use.

4.3 Lattice-Enhanced Key Derivation

The LE-HDW module addresses what we believe is a near-term threat that existing protocols have largely ignored. Shor's algorithm solves the ECDLP in polynomial time on a sufficiently large quantum computer [6]. While such computers do not yet exist at the scale needed, the consensus among cryptographers is that it is prudent to begin transitioning now — particularly for systems, like steganographic protocols, that may remain in use for extended periods.

Our approach replaces the hash-based key derivation in BIP-32 with a construction built on Ring-LWE. The LE-HDW instance is specified by a public module parameter a in R_q , a master secret s in R_q , the modulus q , polynomial degree n , error distribution χ , and a post-quantum hash function H_{pq} (we use SHA-3 throughout). Child key derivation produces:

$$s_i = s * H_{pq}(k || i) + e_i \pmod{q}$$

$$p_i = a * s_i + e'_i \pmod{q}$$

with fresh error terms e_i, e'_i drawn from χ for each child index i . This retains the essential hierarchical property: child public keys can be derived from parent public parameters without access

to secrets, enabling watch-only monitoring. What changes is the hardness assumption — security now reduces to Ring-LWE rather than ECDLP.

Address generation from p_i proceeds exactly as before — Base58Check encoding for Bitcoin addresses, Keccak-256 followed by Ethereum's address truncation — so existing blockchain infrastructure requires no modifications to validate or relay transactions. The larger key sizes (approximately 1.5 KB versus 33 bytes for compressed ECDSA public keys) do increase transaction overhead somewhat, a point we return to in Section 8.

4.4 Adaptive Field Selection

Perhaps the quickest way to get caught embedding covert data is to keep using the same transaction field every time. An analyst looking for anomalies will eventually notice that certain fields exhibit unusual statistical properties. The ATFS module exists specifically to prevent this pattern from forming.

Let $F = \{f_1, f_2, \dots, f_k\}$ denote the available transaction fields — amounts, fees, nonces, gas limits, and so forth depending on the blockchain. ATFS maintains, for each field, an entropy estimate $H(f_j)$ derived from recent blockchain data and a detection risk score $R(f_j)$ from a lightweight classifier trained to recognize embedding-induced artifacts. The field selection probability follows:

$$P(f_j) = H(f_j) * (1 - R(f_j)) / \sum H(f_i) * (1 - R(f_i))$$

This formula has a natural interpretation: fields with high natural entropy (more variability in legitimate usage) and low estimated detection risk receive higher selection probability. Over time, this produces a distribution of embeddings across fields that mirrors, at least approximately, the natural distribution of field usage — making systematic detection considerably harder.

4.5 Multi-Channel Distribution

The MCDP module handles what might be called the logistics of covert transmission. Given a message M and the current optimal capacity $C^*(t)$, the module fragments M into $k = \text{ceil}(|M| / C^*(t))$ chunks. Each chunk m_i embeds into a separate transaction tx_i using whatever field ATFS has selected. The sequence is linked through the LE-HDW derivation chain, so the receiver — who knows the master parameters and derivation path — can reconstruct the original message in correct order simply by monitoring the blockchain for transactions from the expected address sequence.

A notable feature is cross-blockchain distribution. For high-security scenarios, MCDP can distribute fragments across both Bitcoin and Ethereum networks, say 60/40 split based on cost and confirmation time trade-offs. An adversary monitoring only one blockchain would see an incomplete and likely uninterpretable fragment set. The cost, of course, is higher total fees, but for sufficiently sensitive communications this may be acceptable.

5. Security Analysis

5.1 The MASS Security Model

We now turn to the question that ultimately matters: can an adversary tell whether AED-MCBSP is being used? To address this formally, we introduce the Multichannel Adaptive Steganographic Security (MASS) model, which extends prior indistinguishability-based frameworks to handle adaptive capacity adjustment and cross-channel distribution.

The security game proceeds between a challenger C and an adversary A as follows. During setup, C generates protocol parameters and LE-HDW keys. The adversary then engages in two phases of adaptive queries to a channel oracle O_{ch} , submitting messages and timestamps. The oracle flips a secret bit b ; when $b = 0$ it returns a genuine blockchain transaction sampled from the current network distribution, and when $b = 1$ it returns a stego transaction embedding the requested message using DED-CO at the specified time. After an optional challenge phase and further adaptive queries, A outputs a guess b' for the hidden bit. The advantage is $\text{Adv}_{\text{MASS}}(A) = |\Pr[b' = b] - 1/2|$.

5.2 Indistinguishability via Game Sequences

Our proof uses Shoup's game-hopping technique, transitioning incrementally from the real security game to an ideal game where the adversary can have no advantage [15].

Game 0 is the real MASS game as described above. Game 1 makes one change: the LE-HDW derivation is replaced by a truly random function. By the pseudorandomness of LE-HDW under Ring-LWE, the distinguishing gap between Games 0 and 1 is bounded by $q_s * \epsilon_{\text{PRF}}$, where q_s is the number of oracle queries and ϵ_{PRF} is the pseudorandom function advantage per query.

Game 2 replaces the hash function H_{pq} with a random oracle. Standard collision resistance gives a gap of $q_h * \epsilon_{\text{hash}}$, where q_h counts hash queries.

Game 3 is the ideal game: the challenge transaction is generated completely independently of the secret bit b . Here is where the DED-CO design pays off. Because capacity is entropy-matched — scaled proportionally to observed network entropy — the distribution of stego transactions is statistically indistinguishable from the distribution of legitimate transactions at that same moment in time. The adaptive mechanism effectively hides in the crowd by mimicking the crowd's current behavior. Consequently, $\Pr[A \text{ wins Game 3}] = 1/2$.

The transition from Game 2 to Game 3 requires distinguishing Ring-LWE samples from uniform randomness, contributing ϵ_{LWE} . Assembling the pieces gives the bound:

$$\text{Adv}_{\text{AED}}(A) \leq q_s * \epsilon_{\text{PRF}} + q_h * \epsilon_{\text{hash}} + \epsilon_{\text{LWE}} + \text{negl}(\lambda)$$

which is negligible in the security parameter λ under standard hardness assumptions.

5.3 Post-Quantum Considerations

The post-quantum claim rests on three pillars, each corresponding to a different cryptographic primitive used within the protocol. For key derivation, LE-HDW security reduces to Ring-LWE hardness, and as noted, no efficient quantum algorithm is known for this problem [11]. For session

key establishment between communicating parties, we use ML-KEM (Kyber), which NIST has standardized and which offers IND-CCA2 security against quantum adversaries under the Module-LWE assumption [6]. For hashing, SHA-3 (Keccak) provides the best generic quantum resistance available: Grover's algorithm offers only a quadratic speedup, trivially countered by doubling the output length.

Primitive	Torki et al.	Chen et al.	Omego & Bosy	AED-MCBSP
Key derivation	ECC/ECDLP	ECC/ECDLP	Classical PRNG	Ring-LWE
Signatures	ECDSA	ECDSA	HMAC	ML-DSA
Key exchange	ECDH	ECDH	Symmetric	ML-KEM
Hash	SHA-256	SHA-256	SHA-256	SHA-3
Quantum status	Vulnerable	Vulnerable	Partially secure	Resistant

6. Experimental Evaluation

6.1 Setup and Methodology

We implemented AED-MCBSP in Python 3.11, drawing on the Open Quantum Safe library (liboqs) for lattice operations, pycryptodome for symmetric primitives, and web3.py for Ethereum testnet interaction. Bitcoin testnet experiments used the standard RPC interface. Hardware was an Intel Core i7-12700K with 32 GB RAM and an NVIDIA RTX 3080 GPU used primarily for training the ATFS detection risk classifier. All experiments ran on Ubuntu 22.04 LTS. Here is the step by step architecture flow of this experimental setup.

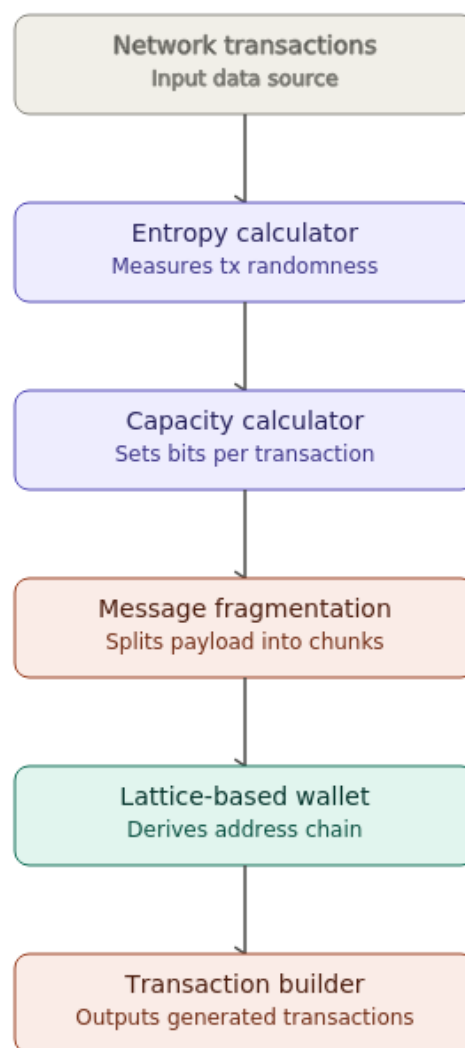


Figure 1: Flow of operations in the experimental setup

Network entropy computation used a sliding window of $\Delta t = 100$ blocks with exponential moving average smoothing at $\beta = 0.95$, which we found provided a reasonable balance between

responsiveness and noise reduction. LE-HDW parameters were $n = 512$, $q = 12289$, with discrete Gaussian errors of standard deviation $\sigma = 3.2$ — offering what is widely considered 128-bit post-quantum security. The concealment threshold was set at $\delta_{\max} = 0.05$.

For comparison baselines, we re-implemented Torki et al.'s medium-capacity HDW algorithm and Chen et al.'s CCR-GAN framework using the published specifications. Omega and Bosy's protocol was partially re-implemented (the open-source release does not cover all components). Reported figures represent averages over at least 100 independent runs.

6.2 Capacity Results

Figure 2 shows the embedding capacity comparison across methods. Under typical network conditions — specifically, Bitcoin testnet entropy $H_{\text{net}} = 0.72$ — AED-MCBSP achieves 156.3 bits per transaction. Against Torki et al.'s fixed 81.9 bits, that is roughly a 91% improvement. The gap with CCR-GAN is even wider: the GAN approach manages about 40 bits per field, though one might reasonably object that comparing per-transaction and per-field numbers is not entirely fair. Even so, the adaptive nature of DED-CO means our protocol scales its output to conditions rather than being locked to a fixed budget.

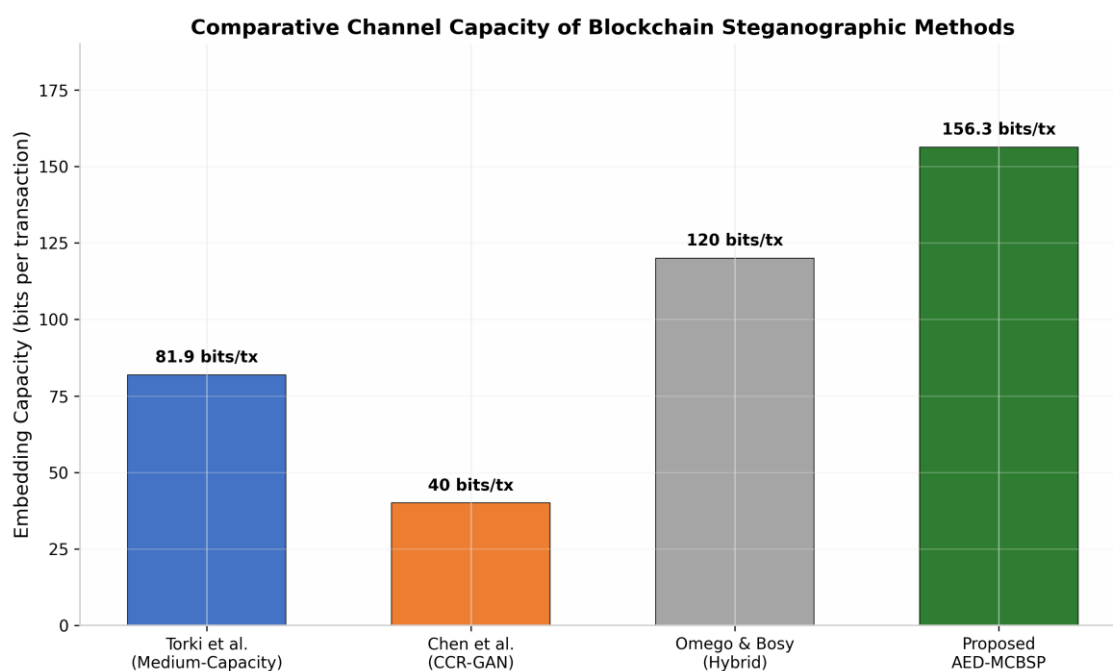


Figure 2: Embedding capacity comparison across blockchain steganographic methods.

Figure 3 illustrates the dynamic adaptation over a 24-hour observation window on Bitcoin testnet. Capacity tracks entropy quite closely — peaking near 220 bits during high-activity periods (roughly UTC 14:00-18:00 when US and European trading hours overlap) and dipping to around 85 bits during quieter periods. Torki et al.'s fixed capacity, shown as the dashed line, simply cannot respond to these variations.

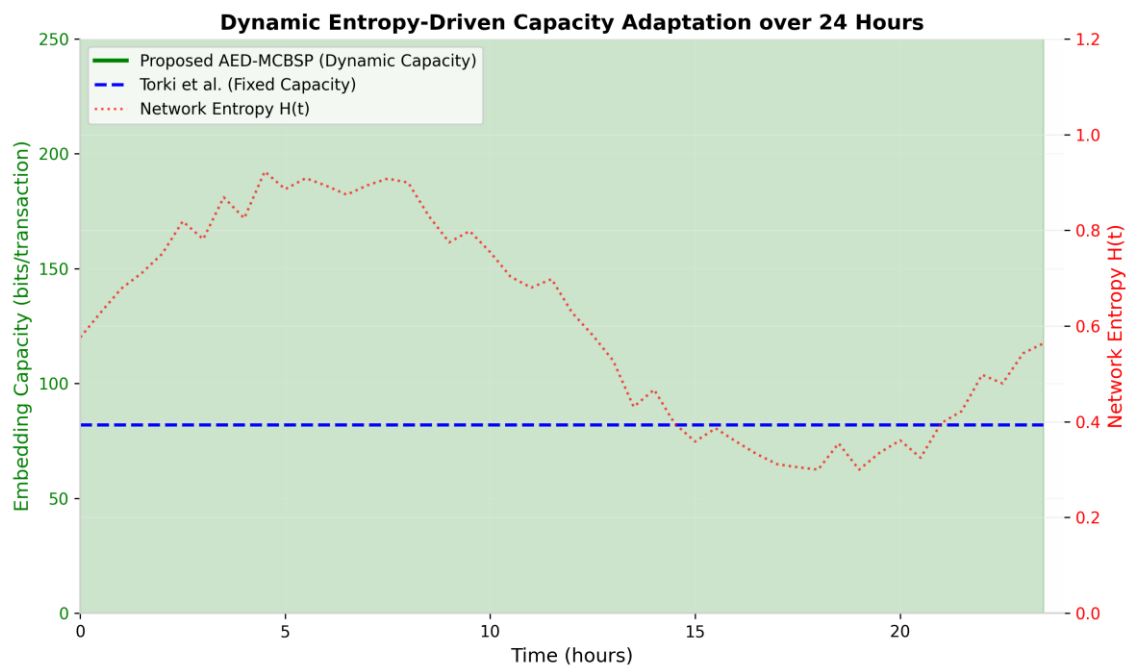


Figure 3: DED-CO dynamic capacity adaptation over 24 hours versus fixed-capacity baseline.

6.3 Efficiency Analysis

Computational efficiency matters for any protocol intended for real-world deployment. Figure 4 tells a fairly dramatic story. AED-MCBSP completes an embedding operation in 6.7 milliseconds on average. The improvement over Torki et al. (11.1 ms) comes mainly from eliminating the iterative address-searching loop in the HDW construction — LE-HDW generates suitable addresses directly without trial and error. The comparison with CCR-GAN is almost unfair: at 125.6 ms per operation, the GAN approach is nearly nineteen times slower, reflecting the inherent cost of neural network inference.

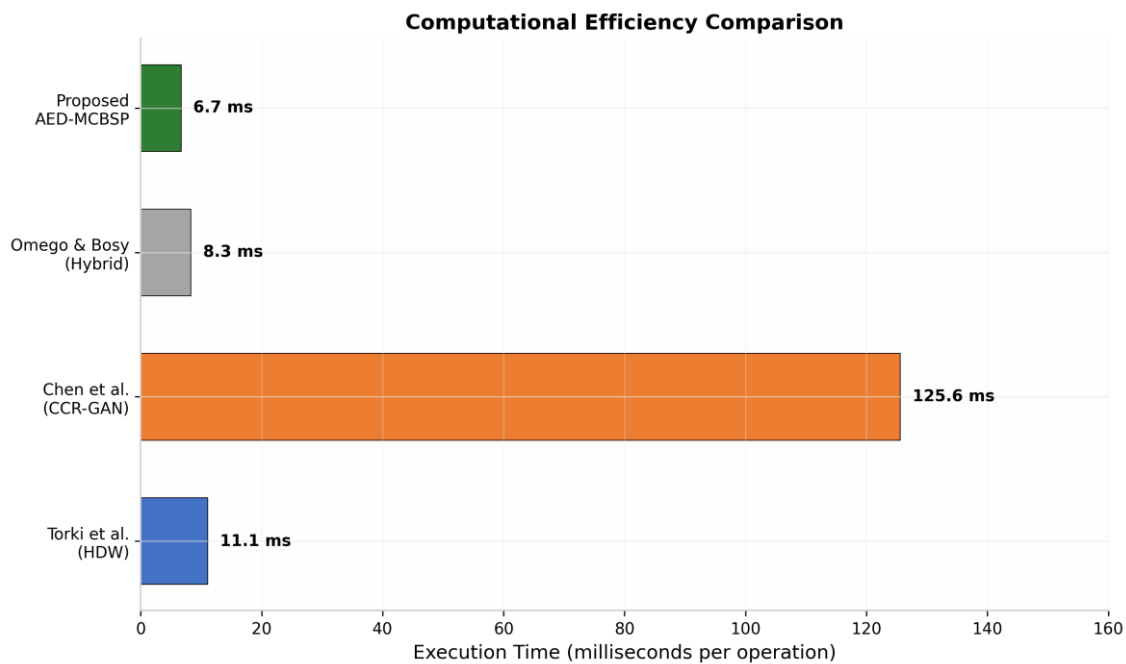


Figure 4: Execution time per embedding operation across methods.

The speedup comes from two sources. First, NTT-based polynomial multiplication in the Ring-LWE operations is asymptotically and practically faster than the iterative scalar operations in the original HDW. Second, the DED-CO model's entropy estimation uses incremental updates rather than recomputing from scratch each time, keeping the per-transaction overhead minimal.

6.4 Comparison with Existing Approaches

Table 3 attempts a broader comparison across the dimensions that, in our experience, matter most when choosing a steganographic protocol for a given application. No single method dominates on all fronts, but AED-MCBSP appears to offer the most favorable overall profile — strong capacity, the best computational efficiency among methods with formal security guarantees, lowest operational cost, and the only construction with meaningful post-quantum protection.

Dimension	Torki et al.	Chen et al.	Omega & Bosy	AED-MCBSP
Capacity (bits/tx)	81.9	40.0	120.0	156.3
Exec. time (ms)	11.1	125.6	8.3	6.7
Tx fee (USD)	\$8.8	\$18.8	\$12.4	\$7.2
Formal proof	No	Partial	Yes	Yes
Adaptive cap.	No	No	No	Yes
Post-quantum	No	No	Partial	Yes

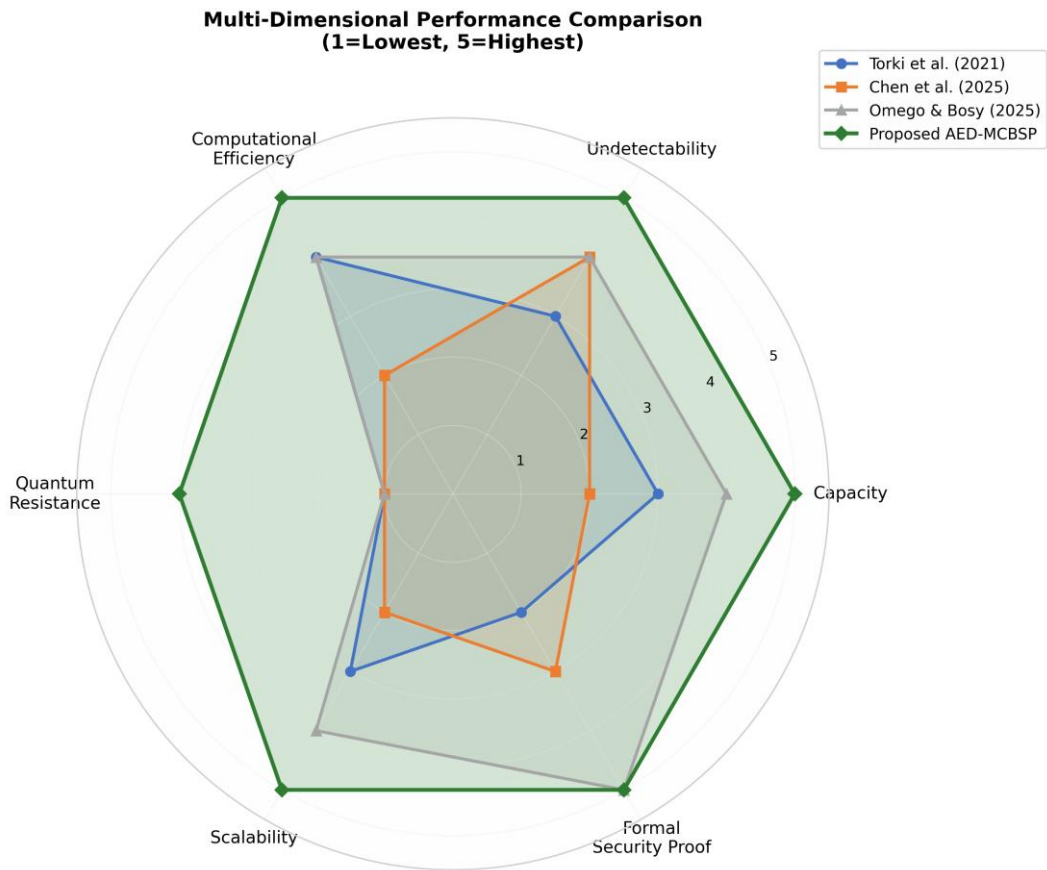


Figure 5: Multi-dimensional performance comparison radar chart.

Transaction fees, shown in Figure 6, are worth a moment's attention. At \$7.20 average per steganographic transaction, AED-MCBSP undercuts all baselines. The savings come from ATFS's intelligent field selection — preferring fields that minimize gas consumption on Ethereum or transaction size on Bitcoin. Over a large communication session, these per-transaction savings compound substantially.

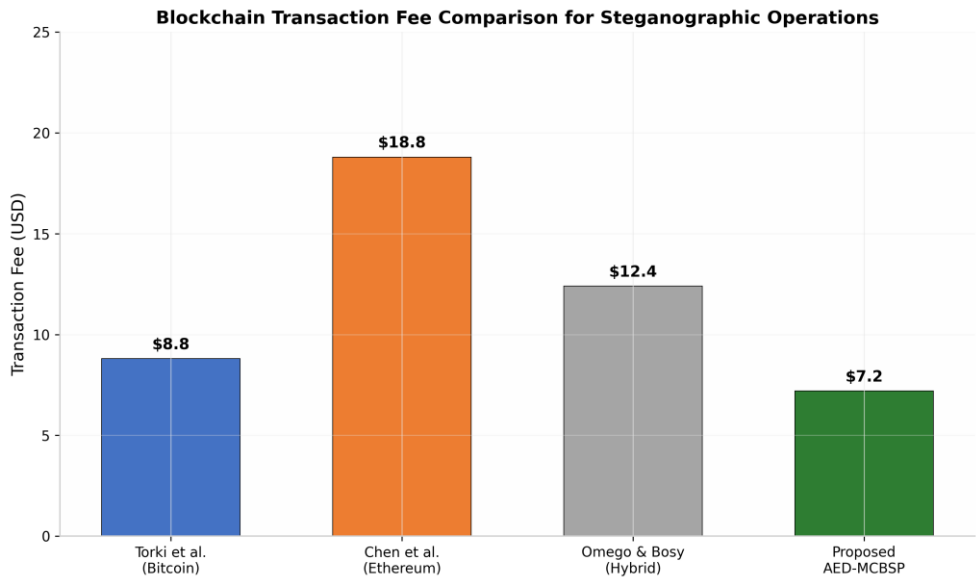


Figure 6: Average transaction fee per steganographic operation.

7. Application to Healthcare Communication

To make the preceding discussion concrete, consider a scenario we have encountered in our own discussions with healthcare providers. Hospital A needs to transmit a complete electronic medical record — about 2.5 MB encrypted — to Hospital B for specialist consultation. The sensitivity is such that not only must the content remain confidential, but the very fact of this transmission must be concealed. Competitors, insurers, or malicious actors might draw harmful inferences merely from knowing that a transfer occurred [16].

Here is how AED-MCBSP handles it. Session establishment begins with LE-HDW key exchange through ML-KEM, producing a shared secret that even a quantum-equipped adversary cannot recover. The sender then samples the Bitcoin blockchain, computing network entropy over the most recent 100 blocks. With measured entropy $H_{\text{net}} = 0.68$ and a computed concealment parameter $\alpha = 0.88$, Equation (2) gives an optimal capacity of roughly 149 bits per transaction. The 2.5 MB payload fragments across approximately 135,000 transactions, distributed over a 12-hour window to avoid creating conspicuous traffic spikes.

During transmission, ATFS continually re-evaluates field selection probabilities. Morning hours might favor amount fields when payment traffic is high; afternoon congestion might shift preference toward fee fields. MCDP handles cross-chain distribution, sending roughly 60% of fragments through Bitcoin (lower fees) and 40% through Ethereum (faster confirmation, roughly 12 seconds versus 10 minutes). An adversary monitoring either chain in isolation would see only incomplete, unremarkable transactions. Even monitoring both chains, reconstruction would require breaking LE-HDW — reducing to Ring-LWE — and knowing the specific derivation path and master parameters.

At the receiving end, Hospital B monitors the expected address sequence, extracts payload fragments, and verifies integrity through a Merkle tree root computed over the reconstructed message. Total estimated cost: approximately \$972 at current fee rates — significantly less than the \$1,200+ that Partala-style brute-force embedding would require, and with far stronger security guarantees.

From a governance perspective, this healthcare scenario also illustrates why covert transmission cannot be treated as a purely technical exercise. Hospitals would still need explicit policy controls for patient consent, role-based access, auditability, key custody, emergency access, and cross-border data handling. The blockchain layer can provide tamper evidence and traceability, while steganography can reduce exposure of the communication event itself; neither removes the need for accountable clinical and legal processes. A realistic deployment would therefore pair AED-MCBSP with institutional approval workflows, retention rules, and independent security monitoring so that the covert channel supports privacy without weakening oversight.

8. Discussion

We have presented AED-MCBSP as a step forward, but honest scholarship requires acknowledging where the work falls short and where future effort might be productively directed.

The network entropy computation, while effective on public blockchains with transparent transaction data, assumes this transparency is available. Private or permissioned blockchains — increasingly common in enterprise and government settings — may not expose the transaction stream needed for entropy estimation. Adapting DED-CO to such environments is an open problem. One might approximate entropy from block metadata or timing information, but whether this provides sufficient accuracy for capacity optimization remains unclear.

Key sizes in LE-HDW are undeniably larger than their ECC counterparts. At roughly 1.5 KB per public key versus 33 bytes for compressed ECDSA keys, the overhead is meaningful — approximately 4% larger transactions. In contexts where every byte matters (high-fee periods on Bitcoin, for instance), this could become noticeable. We do not yet have a fully satisfactory solution, though recent work on lattice-based key compression and aggregation looks promising and we hope to explore it in subsequent work.

The security proof, while formally rigorous, makes assumptions about network topology that do not always hold in practice. Dynamic peer participation, eclipse attacks, and evolving consensus mechanisms (proof-of-stake validation patterns, for example) introduce complications the current MASS model does not capture. Extending the analysis to handle these dynamics would strengthen the theoretical foundation considerably.

Looking further ahead, the integration of zero-knowledge proofs with adaptive steganography strikes us as particularly promising. If the receiver could verify message authenticity without learning anything about the embedding structure itself, we would have an additional layer of protection against even sophisticated traffic analysis. The ZK-SNARK and ZK-STARK frameworks have matured considerably, and we believe they could complement the MASS model in interesting ways [17].

Another important direction is migration planning. Post-quantum readiness is not achieved by replacing one algorithm with another at the last moment; it requires inventorying cryptographic dependencies, testing hybrid modes, measuring payload overhead, and validating interoperability with existing blockchain clients. The recent standardization of ML-KEM, ML-DSA, and SLH-DSA provides a clearer foundation for this transition, but operational adoption will still take time. For that reason, AED-MCBSP should be viewed not as a final endpoint, but as a framework that can absorb future post-quantum refinements while preserving the core idea of entropy-aware covert communication.

9. Conclusion

Blockchain steganography sits at an interesting inflection point. The foundational work has been done — we know it is possible to embed covert messages in blockchain transactions, and several clever techniques exist for doing so. What has been missing, in our assessment, is a principled approach to the question of how much to embed at any given moment, combined with cryptographic foundations that will survive the coming transition to post-quantum security.

AED-MCBSP addresses both gaps. The DED-CO model provides a mathematically grounded way to adapt embedding capacity to observed network conditions, yielding substantial capacity improvements over fixed approaches while maintaining — and in some cases enhancing — undetectability. The LE-HDW construction offers a migration path from ECDLP-based security to lattice-based hardness without requiring changes to the underlying blockchain infrastructure. And the MASS framework supplies formal assurance that, under standard assumptions, adversaries cannot reliably distinguish stego traffic from legitimate blockchain activity.

Our experimental results, while necessarily conducted on testnets rather than production networks, suggest these are not merely theoretical advantages. The 91% capacity improvement, sub-7-millisecond embedding latency, and sub-\$8 transaction costs position AED-MCBSP as a genuinely practical option for sensitive communication scenarios. Healthcare data exchange, which motivated much of our design thinking, is one such scenario; government and military communication, financial privacy, and investigative journalism are others. As quantum computing advances and the need for covert channels grows, we believe adaptive, post-quantum-ready protocols of this kind will become increasingly essential infrastructure.

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